

# Problem Set 6

## Econ 5253 - Spring 2026

Jhinuk Banerji

Due: March 24, 2026

### Question 1 and 2

Completed `git pull origin master` and synchronized fork with the class repository using `git fetch upstream` and `git merge upstream/master`.

### Question 3: Data Cleaning

The dataset used is the World Bank Gender Statistics dataset scraped in Problem Set 5. It contains country-level data on the percentage of women in national parliaments and female labor force participation rate from 2000 to 2023.

The following cleaning steps were performed in `PS6.Banerji.R`:

- **Removed missing values:** Rows where either variable was `NA` were removed using `filter()`, reducing the dataset to 4,060 complete observations.
- **Renamed variables:** The column `date` was renamed to `year`, and the two key variables were shortened to `women_parl` and `female_labor` for clarity.
- **Filtered year range:** Only observations from 2000 to 2023 were kept for a consistent time period.
- **Recoded country names:** Country names were recoded to match the `maps` package conventions for the world map visualization (e.g., “United States” to “USA”).

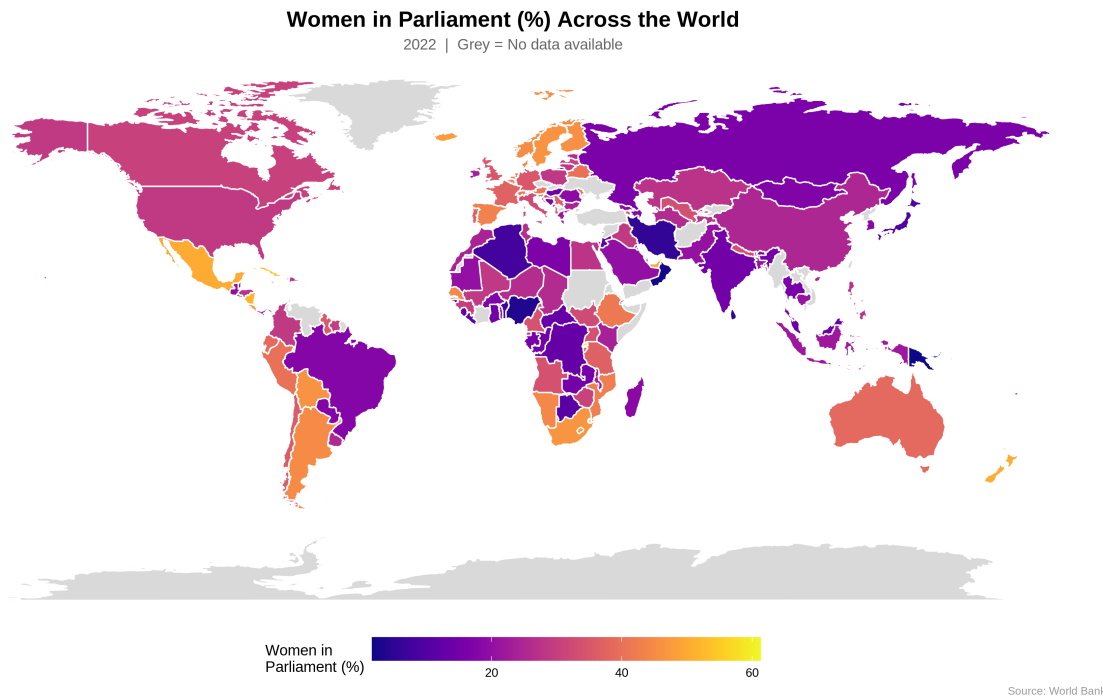


Figure 1: Women in Parliament (%) Across the World, 2022

## Question 4: Visualizations

## Question 5: Explanation of Visualizations

Figure 1 shows a choropleth world map displaying the percentage of women in national parliaments across all countries in 2022. Countries with higher representation appear in brighter colors, while grey indicates missing data. This reveals strong regional patterns: Nordic and some African countries tend to have higher female parliamentary representation, while parts of the Middle East show lower levels.

Figure 2 presents a heatmap of the top 30 countries by average female parliamentary representation from 2000 to 2023. Each row is a country and each column is a year. This is useful for identifying trends over time — Rwanda stands out with dramatically increasing representation over the period.

Figure 3 is a scatter plot showing the relationship between female labor force participation and women in parliament across countries in 2022, with a linear regression line overlaid. The relatively flat regression line suggests these two measures are not strongly correlated, indicating that economic participation alone does not drive political representation.

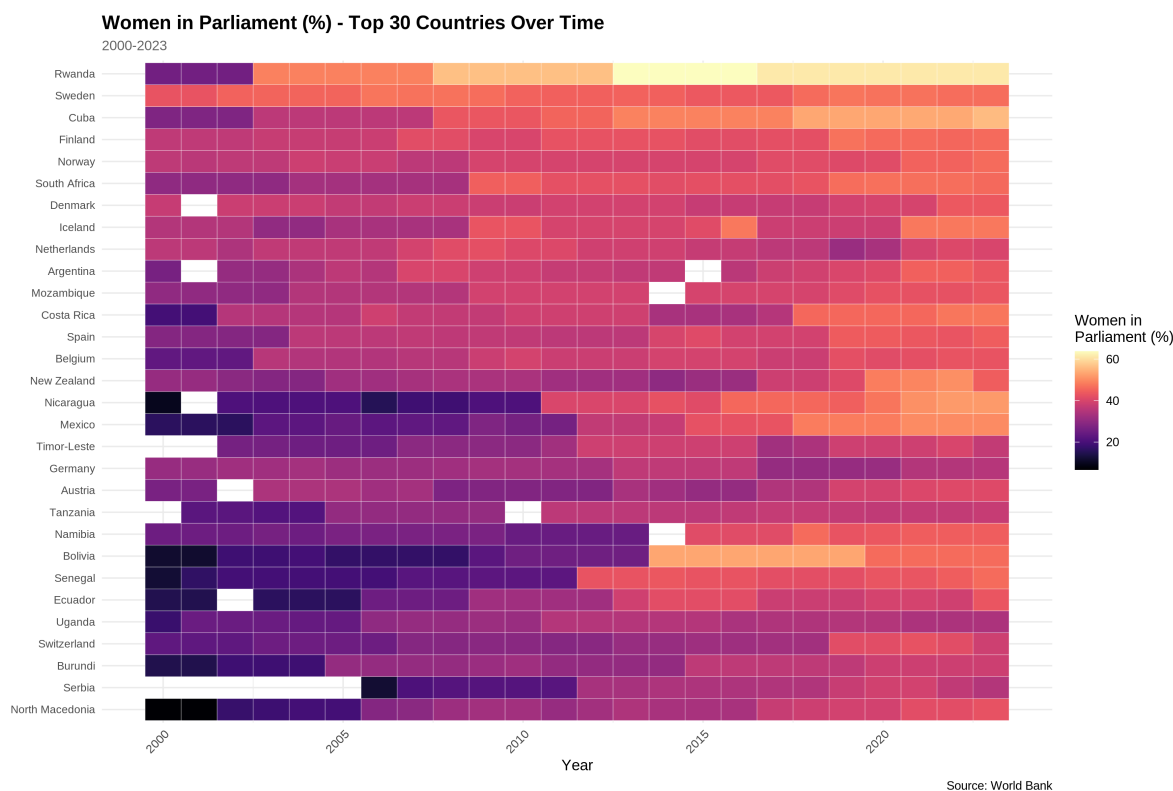


Figure 2: Heatmap of Women in Parliament (%) for Top 30 Countries, 2000–2023

## Questions 6, 7, and 8

Files compiled and transferred to OSCER. All required files are located in `~/DScourseS26/ProblemSets/PS`. Repository synchronized using `git add`, `git commit`, and `git push origin master`.

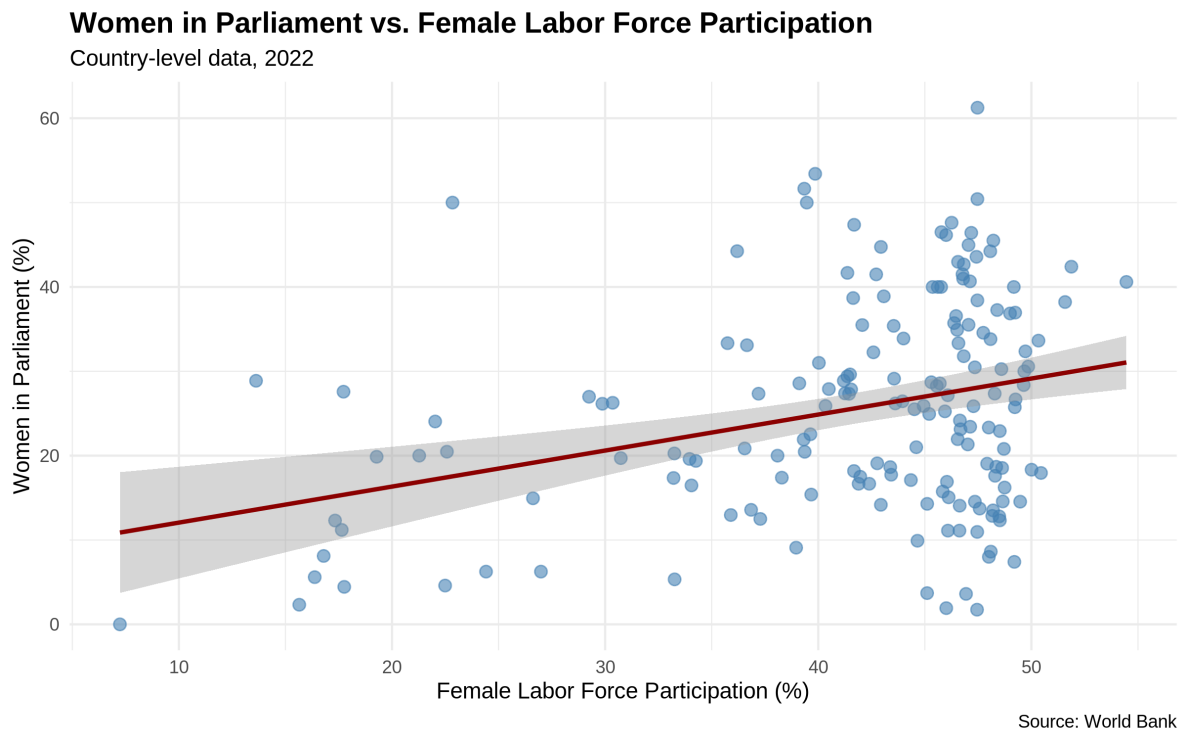


Figure 3: Women in Parliament (%) vs. Female Labor Force Participation (%), 2022